

QLED AND IGZO: The Future Of Television Display Technology



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This article explains how display technologies such as QLED and IGZO are bringing in a revolution in television and other display technologies

Product display screens are output devices that are used to represent output data or information. These are used in almost all electronic devices, such as mobile phones, televisions, refrigerators and computers. A few years ago, cathode ray tubes (CRTs) were used as displays in televisions. Researchers explored this field and, with the help of different technologies, many other types of displays were invented; for instance, light emitting diodes (LEDs), liquid crystal displays (LCDs) and now quantum dot light emitting diodes (QLEDs) and indium gallium zinc oxygen (IGZO). The displays have become bigger as well as thinner because of these technologies.

At present, televisions come with full high-definition (HD) resolution and a thickness of approximately 25mm along with Wi-Fi support, voice recognition feature, 3D sound system and other attractive features. This article focuses on the latest QLED and IGZO display technologies. These technologies are responsible for bringing in a revolution in television and other displays such as those in mobile phones. Quantum dots are the key elements of these technologies. Hence, we need to analyse the quantum dot and its effect first before discussing QLED.

Introduction to quantum dot

Quantum dots are tiny semiconductor neon-scale crystals, about 2nm to 10nm in size. These can be precisely controlled with respect to size, energy level and emission wavelength for various applications like energy-efficient displays, solar cells and biological medical research. According to Quantum Physics, quantum dots emit colour according to the size of the quantum. Bigger the quantum dot, larger the wavelength. Fig. 1 shows the graph detailing wavelength versus fluorescence, which depends on the size of the quantum dot and the energy gap of the material.

Quantum dots can be created using different methods, but generally these are made by chemical reactions in a solution. The chemical reaction is done over 200 + °C temperature. Many different semiconductor material combinations such as cadmium selenide (CdSe), cadmium sulphide (CdS), lead selenide (PbSe), lead sulphide (PbS), indium arsenide (InAs) and indium phosphide (InP) are used to make quantum dots. These absorb light and release energy in terms of different colours of light according to the band gap of the material, as shown in Fig. 1.

Quantum dots, till recently, were generally made of CdS or CdSe. These particles are hazardous to the environment, and during in-vitro degradation, Cd²⁺ ions may get discharged. Samsung uses cadmium-free quantum dot technology, and at present quantum dots use indium (Id). This technology can produce numerous colour combinations (approximately one billion) that provide better light efficiency.

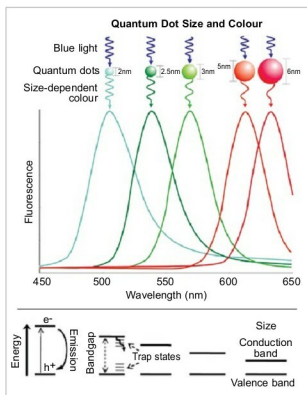


Fig. 1: Quantum dot colour emission spectrum (Source: <http://physicsopenlab.org>)